

A Finite-Skeleton Construction of Berkovich Holography

Yifan Yao

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Abstract

The standard p -adic AdS/CFT model places a scalar field on the Bruhat–Tits tree and extracts boundary correlation functions on $\mathbb{P}^1(K)$, or on an affine chart of a non-Archimedean local field K . Since the Bruhat–Tits tree embeds in the Berkovich projective line, one may ask whether the tree model has a natural enlargement in which the bulk is the hyperbolic Berkovich line $\mathbb{H}_K = \mathbb{P}_K^{1,\text{an}} \setminus \mathbb{P}^1(K)$.

The main point is negative before it is positive: the full Berkovich line is not a locally finite regular graph, so the Bruhat–Tits graph action, counting measure, regular branching number, and zeta-normalized massive Green functions do not extend canonically. A rigorous formulation must therefore begin at finite cutoff. We define scalar holography on finite Berkovich skeleta, prove the corresponding Dirichlet problem, construct the Dirichlet-to-Neumann boundary quadratic form, and formulate finite-skeleton tree-level Witten diagrams. We then explain how the Bruhat–Tits model is recovered as a homogeneous locally finite specialization and how the Hsia kernel and Berkovich Green functions supply natural, but not automatic, continuum candidates.

The article is written as a research note rather than as a completed theorem of continuum Berkovich holography. Every assertion that is proved at finite cutoff is separated from claims that require an exhaustion, counterterms, and convergence theorems.

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1 Purpose and correction of the naive proposal

Let K be a complete non-Archimedean field. The geometric temptation is simple: replace the Bruhat–Tits tree T_K by the hyperbolic Berkovich line

$$\mathbb{H}_K := \mathbb{P}_K^{1,\text{an}} \setminus \mathbb{P}^1(K),$$

regard the classical points $\mathbb{P}^1(K)$ as the conformal boundary, and use Berkovich potential theory in place of graph potential theory. This is conceptually correct, but it is not by itself a complete theory. The missing choices are essential.

Warning 1.1 (The naive continuum statement is false as stated). The phrase “the bulk is $\mathbb{H}_K$ and the action is the massive scalar action on $\mathbb{H}_K$ ” does not define a mathematical object until one specifies:

- (i) a class of admissible finite or locally finite subtrees,
- (ii) a bulk measure or volume functional,
- (iii) a boundary measure for sources,
- (iv) a Laplacian domain and boundary condition,
- (v) a renormalization scheme for approaching type-I boundary points.

The Bruhat–Tits model hides these choices because the regular tree supplies a canonical counting measure, a canonical graph Laplacian, and a homogeneous boundary Haar measure. The full Berkovich projective line does not supply these data uniquely.

The aim of this note is to build a version that is correct at finite stage and that has a well-defined route to a continuum limit. The guiding principle is:

Berkovich AdS/CFT should first be constructed on finite Berkovich skeleta.

Only after this finite construction is made should one take an exhaustion limit, and the limit should be regarded as part of the definition of the theory.

1.1 Relation to the Bruhat–Tits construction

For a non-Archimedean local field K with residue cardinality q , the Bruhat–Tits tree of $\mathrm{PGL}_2(K)$ embeds in $\mathbb{P}_K^{1,\mathrm{an}}$ as the tree of type-II discs with radii in $|K^\times|$, together with its natural adjacency structure. The scalar model of p -adic AdS/CFT uses this regular tree as the bulk and $\mathbb{P}^1(K)$ as the boundary. On a $(q+1)$ -regular tree, the graph Laplacian and the counting measure produce the familiar propagators and local-zeta normalization factors.

The finite-skeleton Berkovich theory below recovers the Bruhat–Tits theory when the chosen skeleta are finite truncations of the Bruhat–Tits tree and the bulk measure is the counting measure on vertices. Thus the Bruhat–Tits theory is not discarded; it is the homogeneous locally compact special case.

1.2 Scope and relation to existing literature

This note does not claim originality for the ingredients separately. Bruhat–Tits-tree p -adic AdS/CFT, including scalar propagators and contact diagrams, was developed in [6] and subsequent work. Non-Archimedean strings and the role of Bruhat–Tits trees go back to [9, 10]. Potential theory on $\mathbb{P}_K^{1,\mathrm{an}}$ is the subject of [2, 3, 13], and the canonical measures used in non-Archimedean dynamics are developed in, for example, [4]. Analysis on finite metric graphs is standard in the quantum and metric graph literature, see for instance [11, 12]. Berkovich spaces have also appeared in physics-adjacent discussions of p -adic-to-Archimedean renormalization flow, for example [8].

The contribution of the present note is only the following conservative synthesis: a Berkovich enlargement of p -adic holography should be treated first as a finite-skeleton Dirichlet-to-Neumann construction, and any continuum Berkovich theory should be defined as a renormalized exhaustion limit. This avoids the misleading statement that there is a unique canonical scalar AdS action on the whole hyperbolic Berkovich line.

2 Berkovich preliminaries

This section fixes notation. We use only standard facts from Berkovich analytic geometry and non-Archimedean potential theory.

2.1 The Berkovich projective line

Let $(K, |\cdot|)$ be a complete non-Archimedean field. The Berkovich affine line $\mathbb{A}_K^{1,\mathrm{an}}$ consists of multiplicative seminorms on $K[T]$ extending the absolute value on K . The Berkovich projective line $\mathbb{P}_K^{1,\mathrm{an}}$ is obtained by adding the point at infinity and gluing the two standard affine charts.

In an affine chart, a large and geometrically important class of points is represented by closed discs

$$D(a, r) = \{z \in K : |z - a| \leq r\}, \quad a \in K, \quad r \geq 0,$$

with corresponding seminorm

$$\zeta_{a,r}(f) = \sup_{z \in D(a,r)} |f(z)|.$$

When $r = 0$, the point $\zeta_{a,0}$ is the classical point $a \in K$. Points with $r > 0$ are non-classical bulk points. The Gauss point is

$$\zeta_G := \zeta_{0,1}.$$

Definition 2.1 (Hyperbolic Berkovich line). The hyperbolic Berkovich line is

$$\mathbb{H}_K := \mathbb{P}_K^{1,\text{an}} \setminus \mathbb{P}^1(K).$$

It is the proposed Berkovich bulk. The type-I points $\mathbb{P}^1(K)$ form the proposed boundary.

2.2 Tree structure and branch points

The space $\mathbb{P}_K^{1,\text{an}}$ is uniquely path connected. For any two points X, Y , there is a unique arc $[X, Y]$. For three points X, Y, Z , the intersection

$$[X, Y] \cap [Y, Z] \cap [Z, X]$$

consists of exactly one point; we denote it by

$$X \vee Y \vee Z.$$

For $x, y, z \in \mathbb{P}^1(K)$ this is the Berkovich branch point at which the three boundary geodesics meet.

2.3 Diameter, Hsia kernel, and path distance

In the affine chart containing K , define

$$\text{diam}(\zeta_{a,r}) := r.$$

For $X = \zeta_{a,r}$ and $x \in K$, define the affine Hsia-type expression

$$\delta(X, x) := \max\{r, |a - x|\}.$$

More invariantly, after choosing a reference point such as the Gauss point, one has a normalized Hsia kernel

$$[X, Y]_{\zeta_G}$$

which extends the chordal distance on classical points and is a basic object of Berkovich potential theory. In the affine chart and away from infinity, the formula above is the concrete expression we shall need.

For two disc points $X = \zeta_{a,r}$ and $Y = \zeta_{b,s}$, define

$$R(X, Y) := \max\{r, s, |a - b|\}.$$

The path distance on the hyperbolic line is, up to a harmless normalization convention,

$$\rho(X, Y) = 2 \log R(X, Y) - \log r - \log s. \quad (1)$$

This is the logarithmic length of the path from X to Y through their least common ancestor disc. The formula is meaningful for $r, s > 0$.

Remark 2.2. The Bruhat–Tits tree corresponds to restricting r and s to discrete radii, usually $r, s \in |K^\times|$, and then normalizing ρ so that adjacent vertices have length one. Thus graph distance is a discretized Berkovich path distance.

3 Finite Berkovich skeleta

The finite-skeleton construction is the rigorous core of the theory.

3.1 Finite metric subtrees

Definition 3.1 (Finite cutoff Berkovich skeleton). A finite cutoff Berkovich skeleton is a compact connected subset

$$\Gamma \subset \mathbb{H}_K$$

which is a finite union of nondegenerate Berkovich segments, has only finitely many branch points, and has finite total length for the hyperbolic path metric. Equivalently, for the purposes of this note, one may take

$$\Gamma = \text{Hull}(A) \cap \{X \in \mathbb{H}_K : \rho(X, X_0) \leq L\}$$

where $A \subset \mathbb{P}^1(K)$ is finite, $X_0 \in \mathbb{H}_K$ is a base point, and $L < \infty$. The intersection with the finite ball is essential: a segment from a type-I point to the interior has infinite hyperbolic length.

Let $V(\Gamma)$ be a finite vertex set containing all branch points and endpoints of Γ . Then each connected component of $\Gamma \setminus V(\Gamma)$ is an open interval. We call the closures of these intervals edges and denote the edge set by $E(\Gamma)$. Each edge e is isometric to a compact interval $[0, \ell_e]$.

Definition 3.2 (Boundary of a finite skeleton). A boundary choice for Γ is a nonempty finite subset

$$B \subset V(\Gamma)$$

containing all leaves of Γ at which boundary data are to be imposed. The interior vertices are

$$V^\circ(\Gamma) := V(\Gamma) \setminus B.$$

This definition is deliberately flexible. In a cutoff construction, B consists of vertices close to, but not equal to, the type-I boundary. In a purely finite problem, B may be any chosen set of external vertices.

Warning 3.3 (Do not put type-I endpoints in the finite metric action). The topological hull of finitely many type-I points is compact in the Berkovich topology, but its hyperbolic edges running to type-I points have infinite length. The finite action below is therefore imposed on truncated skeleta whose boundary vertices lie in $\mathbb{H}_K$. The type-I boundary is reached only after an exhaustion and renormalization limit.

3.2 Function spaces and energy

Let $H^1(\Gamma)$ be the space of continuous functions $\phi : \Gamma \rightarrow \mathbb{R}$ whose restriction to each edge lies in the ordinary Sobolev space $H^1([0, \ell_e])$. Define

$$H_0^1(\Gamma, B) := \{\phi \in H^1(\Gamma) : \phi|_B = 0\}.$$

Fix a positive finite Borel measure ω_Γ on Γ . This measure is part of the data of the model. The simplest metric-tree choice is $d\omega_\Gamma = ds$, length measure. The Bruhat–Tits graph model is a different, discrete specialization: one first chooses a vertex set and then uses counting measure on vertices together with the graph Dirichlet energy. Thus metric-edge and graph-vertex models are parallel finite-cutoff realizations rather than literally the same measure-theoretic object.

Definition 3.4 (Massive Dirichlet form). For $m^2 \geq 0$, define

$$\mathcal{E}_{\Gamma,m}(\phi, \psi) := \int_{\Gamma} \phi'(s) \psi'(s) ds + m^2 \int_{\Gamma} \phi \psi d\omega_{\Gamma}. \quad (2)$$

The associated free scalar action is

$$S_{\Gamma}^{(2)}[\phi] := \frac{\eta}{2} \mathcal{E}_{\Gamma,m}(\phi, \phi). \quad (3)$$

Remark 3.5. If one wants tachyonic or Breitenlohner–Freedman-type masses, then m^2 may be negative, but coercivity must be checked separately. For the rigorous construction in this note we assume $m^2 \geq 0$ unless explicitly stated otherwise.

3.3 Dirichlet problem

Let $f : B \rightarrow \mathbb{R}$ be boundary data. A function $\phi \in H^1(\Gamma)$ is said to have trace f if $\phi(b) = f(b)$ for all $b \in B$.

Theorem 3.6 (Finite-skeleton Dirichlet problem). *Assume $m^2 \geq 0$ and that each connected component of Γ meets B . For every boundary datum $f : B \rightarrow \mathbb{R}$, there exists a unique minimizer*

$$H_{\Gamma,m}f \in H^1(\Gamma)$$

of $S_{\Gamma}^{(2)}$ among functions with trace f . Equivalently,

$$\mathcal{E}_{\Gamma,m}(H_{\Gamma,m}f, u) = 0 \quad \text{for all } u \in H_0^1(\Gamma, B).$$

Proof. Choose any $F \in H^1(\Gamma)$ with trace f . Then every competitor is of the form $F + u$ with $u \in H_0^1(\Gamma, B)$. It is enough to minimize the strictly convex quadratic functional

$$u \mapsto \frac{\eta}{2} \mathcal{E}_{\Gamma,m}(F + u, F + u)$$

on $H_0^1(\Gamma, B)$. The form $\mathcal{E}_{\Gamma,m}$ is continuous. It is coercive on $H_0^1(\Gamma, B)$ because the derivative term controls the H^1 norm by the Poincaré inequality on each compact tree component meeting the Dirichlet boundary; the nonnegative mass term only improves coercivity. The Lax–Milgram theorem gives a unique critical point, and strict convexity gives uniqueness of the minimizer. The Euler–Lagrange condition is precisely the displayed weak equation. $\square$

Definition 3.7 (Poisson operator). The linear map

$$H_{\Gamma,m} : \mathbb{R}^B \longrightarrow H^1(\Gamma), \quad f \longmapsto H_{\Gamma,m}f,$$

is the finite-skeleton massive Poisson operator.

For each $b \in B$, let $\mathbf{1}_b$ denote the boundary function equal to 1 at b and 0 at the other boundary points. The functions

$$K_{\Gamma,m}(X; b) := (H_{\Gamma,m} \mathbf{1}_b)(X)$$

are the finite-skeleton bulk-to-boundary kernels.

4 Dirichlet-to-Neumann holography on a finite skeleton

The free holographic boundary action is the Dirichlet energy of the harmonic extension.

4.1 Boundary effective action

Definition 4.1 (Dirichlet-to-Neumann form). The finite-skeleton Dirichlet-to-Neumann quadratic form is

$$\Lambda_{\Gamma,m}(f, g) := \mathcal{E}_{\Gamma,m}(H_{\Gamma,m}f, H_{\Gamma,m}g). \quad (4)$$

Equivalently, after choosing the standard basis of $\mathbb{R}^B$, it is a symmetric matrix $\Lambda_{\Gamma,m}(b, b')$.

Proposition 4.2 (On-shell action). *The on-shell free action with boundary source f is*

$$S_{\Gamma,\text{cl}}^{(2)}[f] = \frac{\eta}{2} \Lambda_{\Gamma,m}(f, f).$$

The form $\Lambda_{\Gamma,m}$ is symmetric and positive semidefinite. If $m^2 > 0$ and ω_Γ has support meeting every nonempty open edge, it is positive definite. If $m = 0$, its kernel consists of boundary data that extend to constants on connected components; for connected Γ this is the usual constant zero mode.

Proof. The formula follows immediately from the definition of $S_\Gamma^{(2)}$ and the Poisson operator. Symmetry is inherited from $\mathcal{E}_{\Gamma,m}$. Positivity follows from $\Lambda_{\Gamma,m}(f, f) = \mathcal{E}_{\Gamma,m}(Hf, Hf) \geq 0$. If this vanishes, then $(Hf)' = 0$ and, if $m^2 > 0$ and ω_Γ has support meeting every open edge, also $Hf = 0$ on the support of ω_Γ ; by continuity and connectedness this gives $f = 0$. In the massless case, constants have zero energy, giving the stated zero mode. $\square$

4.2 Finite-skeleton GKP–Witten prescription

The finite-skeleton generating functional is defined by

$$-\log Z_\Gamma[f] := S_{\Gamma,\text{cl}}[f]. \quad (5)$$

For the free theory,

$$-\log Z_\Gamma[f] = \frac{\eta}{2} \Lambda_{\Gamma,m}(f, f).$$

Thus the connected two-point kernel, with the convention $-\log \langle e^{\sum_b f_b O_b} \rangle = S_{\text{cl}}$, is

$$\langle O_b O_{b'} \rangle_{\Gamma,c} = -\eta \Lambda_{\Gamma,m}(b, b'). \quad (6)$$

Many Euclidean field theory conventions couple sources as $e^{-\sum_b f_b O_b}$ or insert an additional sign in the definition of the generating functional; then the displayed sign is reversed. The convention-independent datum is the Dirichlet-to-Neumann form $\Lambda_{\Gamma,m}$.

Remark 4.3 (Why this is the correct finite theory). No global Berkovich measure or continuum Laplacian is needed to make (5) meaningful. The finite skeleton is a compact metric graph, the Dirichlet problem is well-posed, and the boundary effective action is a rigorously defined quadratic form.

5 Interactions and finite Witten diagrams

Let

$$V(\phi) = \sum_{k \geq 3} \frac{\lambda_k}{k!} \phi^k$$

be a polynomial interaction. On a finite skeleton define

$$S_\Gamma[\phi] = S_\Gamma^{(2)}[\phi] + \eta \int_\Gamma V(\phi) d\omega_\Gamma. \quad (7)$$

For sufficiently small couplings, or formally as a perturbation series, the classical solution with boundary data f may be expanded around $H_{\Gamma,m}f$.

Proposition 5.1 (Tree-level contact diagrams). *The leading contribution of the k -point contact interaction $\eta\lambda_k\phi^k/k!$ to the connected boundary k -point function is*

$$\langle O_{b_1} \cdots O_{b_k} \rangle_{\Gamma,c}^{\text{contact}} = -\eta\lambda_k \int_\Gamma \prod_{i=1}^k K_{\Gamma,m}(X; b_i) d\omega_\Gamma(X), \quad (8)$$

up to the same source-sign convention as in (6).

Proof. To first order in λ_k , it is enough to insert the free classical solution

$$\phi_f(X) = \sum_{b \in B} f(b) K_{\Gamma,m}(X; b)$$

into the interaction term. The coefficient of $f(b_1) \cdots f(b_k)$ in $-S_\Gamma[\phi_f]$ is exactly the right-hand side of (8). Terms arising from the first-order correction to the classical solution vanish at this order because the free solution is stationary for the quadratic action. $\square$

Remark 5.2. Exchange diagrams are obtained by inserting the finite-skeleton bulk Green function of the corresponding positive operator with Dirichlet boundary condition: on a metric skeleton this is the self-adjoint operator associated with (2), and on a graph skeleton it is the graph operator (9) plus mass. The important point is that all diagrammatic integrals or sums are finite at finite cutoff.

6 Recovery of the Bruhat–Tits model

We now check that the finite-skeleton formalism recovers the ordinary p -adic AdS/CFT model when specialized to the regular Bruhat–Tits tree.

6.1 The regular tree and its graph Laplacian

Let $K/\mathbb{Q}_p$ be a finite extension with residue cardinality $q = p^n$ in the unramified case considered in many physics applications. Let T_q be the $(q+1)$ -regular Bruhat–Tits tree. For a function ϕ on vertices, define the graph Laplacian

$$(\Delta_T \phi)(a) = \sum_{b \sim a} (\phi(a) - \phi(b)). \quad (9)$$

The free graph action is

$$S_T^{(2)}[\phi] = \frac{\eta}{2} \sum_{a \sim b} (\phi(a) - \phi(b))^2 + \frac{\eta m_p^2}{2} \sum_a \phi(a)^2. \quad (10)$$

This is the discrete graph analogue of (3); it should not be confused with the length-measure metric-graph action on the continuum edges.

6.2 Mass-dimension relation

Fix a geodesic ray and let $r = d(a, a_0)$ be graph distance from a base vertex. Consider the radial ansatz

$$f(r) = p^{-\Delta r}.$$

Away from the base point, the regular tree has one neighbor at distance $r - 1$ and q neighbors at distance $r + 1$. Hence

$$\begin{aligned} (\Delta_T f)(r) &= (q + 1)f(r) - f(r - 1) - qf(r + 1) \\ &= (q + 1 - p^\Delta - qp^{-\Delta})f(r). \end{aligned}$$

The equation

$$(\Delta_T + m_p^2)f = 0$$

therefore gives

$$m_p^2 = p^\Delta + p^{n-\Delta} - p^n - 1. \quad (11)$$

Using

$$\zeta_p(s) = \frac{1}{1 - p^{-s}},$$

this is equivalent to

$$m_p^2 = -\frac{1}{\zeta_p(\Delta - n)\zeta_p(-\Delta)}. \quad (12)$$

This is the usual Bruhat–Tits mass-dimension relation.

Warning 6.1. Formula (12) is a theorem for the homogeneous regular tree. It is not a canonical formula on the full Berkovich line, because the full Berkovich line has no uniform branching number $q + 1$.

6.3 Bulk-to-boundary kernel on the tree

Let a Bruhat–Tits bulk vertex be written in coordinates (z_0, z) , where $z_0 = p^\omega$ records depth and z records the residue class or boundary center. Define

$$|(z_0, z - x)|_s := \max\{|z_0|_p, |z - x|_K\}.$$

The normalized tree bulk-to-boundary kernel is

$$K_\Delta(z_0, z; x) = C_{p,\Delta} \frac{|z_0|_p^\Delta}{|(z_0, z - x)|_s^{2\Delta}}, \quad (13)$$

where, in the standard normalization used in the physics literature,

$$C_{p,\Delta} = \frac{\zeta_p(2\Delta)}{\zeta_p(2\Delta - n)}. \quad (14)$$

This is precisely the restriction of the affine Berkovich expression

$$\mathcal{K}_\Delta(\zeta_{z,r}; x) = C_\Delta \frac{r^\Delta}{\max\{r, |z - x|\}^{2\Delta}} \quad (15)$$

to the discrete radii $r = |z_0|_p$ and the Bruhat–Tits skeleton.

Proposition 6.2 (Skeleton restriction of the Hsia kernel ansatz). *On the Bruhat–Tits skeleton, the candidate Berkovich kernel (15) agrees with the standard tree bulk-to-boundary propagator (13), up to the normalization constant C_Δ .*

Proof. A Bruhat–Tits vertex represented by the disc $D(z, r)$ corresponds to the Berkovich point $\zeta_{z, r}$. For a classical point $x \in K$, the smallest disc containing both $D(z, r)$ and x has radius $\max\{r, |z - x|\}$. Therefore the affine Hsia expression is exactly the supremum norm $|(z_0, z - x)|_s$ after setting $r = |z_0|_p$. Substitution into (15) gives (13). $\square$

6.4 Tree contact diagrams

On the regular tree, the finite-skeleton contact diagram (8) becomes a sum over vertices. For three identical scalar legs one obtains, after removing the cutoff and using the normalized kernel,

$$\langle O(x_1)O(x_2)O(x_3) \rangle = -\eta g_3 \frac{\zeta_p(\Delta)^3 \zeta_p(3\Delta - n)}{\zeta_p(2\Delta - n)^3} \frac{1}{|x_{12}x_{23}x_{13}|_K^\Delta}, \quad (16)$$

up to the normalization convention for the field and the source. The important structural feature is the tree-geometric identity: three boundary points determine a unique branch point, and the sum over the bulk decomposes into geometric series along the branches.

Similarly, four-point contact diagrams reduce to a function of the cross-ratio length

$$\ell_{12|34} = -\log_p \left| \frac{x_{12}x_{34}}{x_{13}x_{24}} \right|_K,$$

in the channel where this expression is nonnegative. This is a consequence of the fact that the overlap of the geodesics $[x_1, x_2]$ and $[x_3, x_4]$ is a finite segment whose length is the valuation of the cross-ratio.

7 The full Berkovich theory as an exhaustion problem

We now formulate the full theory. The definition below intentionally includes the exhaustion and renormalization data.

7.1 Admissible exhaustions

Definition 7.1 (Berkovich holographic datum). A Berkovich holographic datum over K is a tuple

$$\mathfrak{D} = (\mu, \nu, \omega, \{\Gamma_j, B_j, r_j\}_{j \geq 1}, \mathcal{R}),$$

where:

- (i) μ is a reference probability measure on $\mathbb{P}_K^{1, \text{an}}$ with continuous potential;
- (ii) ν is a boundary source measure on a chosen boundary subset $\mathcal{B} \subseteq \mathbb{P}^1(K)$;
- (iii) $\Gamma_j \subset \mathbb{H}_K$ are finite Berkovich skeleta with boundary sets B_j ;
- (iv) $r_j : B_j \rightarrow \mathcal{B}$ is a projection sending each cutoff boundary vertex to the type-I boundary point it approximates;
- (v) ω_j is a positive finite bulk measure on Γ_j ;

(vi) $\mathcal{R}$ is a prescription for subtracting local counterterms and applying leg factors as $j \rightarrow \infty$.

A typical choice is to take finite sets $A_j \subset \mathbb{P}^1(K)$ that become dense in the support of ν and to define a truncated hull

$$\Gamma_j = \text{Hull}(A_j \cup \{\zeta_G\}) \cap \{X \in \mathbb{H}_K : \rho(X, \zeta_G) \leq L_j\},$$

with cutoff leaves B_j lying at distance L_j along the rays toward A_j . The exact definition depends on the field and the boundary problem. For a local field, Bruhat–Tits balls give a canonical homogeneous choice. For $\mathbb{C}_p$, no locally compact translation-invariant Haar measure exists on the additive boundary, so ν must be chosen separately.

7.2 Renormalized generating functional

For a boundary source ϕ_0 on $\mathcal{B}$, define a discretized source $\phi_{0,j} : B_j \rightarrow \mathbb{R}$ by averaging or pulling back along r_j . Let $S_{\Gamma_j, \text{cl}}[\phi_{0,j}]$ be the finite-skeleton on-shell action. The renormalized generating functional is then defined, when the limit exists, by

$$-\log Z_{\mathfrak{D}}[\phi_0] := \lim_{j \rightarrow \infty} \mathcal{R}_j(S_{\Gamma_j, \text{cl}}[\phi_{0,j}]). \quad (17)$$

Here $\mathcal{R}_j$ may include:

- (i) multiplication by leg factors such as $\text{diam}(X_b)^{d-\Delta}$ when the boundary measure has a well-defined scaling dimension d ,
- (ii) subtraction of local divergent terms supported at coincident boundary cells,
- (iii) normalization by the measure of boundary cells,
- (iv) removal of zero modes in the massless case.

Definition 7.2 (Berkovich AdS/CFT, rigorous version). A scalar Berkovich AdS/CFT model is a Berkovich holographic datum $\mathfrak{D}$ together with a choice of finite-skeleton actions S_{Γ_j} such that the renormalized limit (17) exists on a specified class of boundary sources. Boundary correlators are the functional derivatives of $\log Z_{\mathfrak{D}}$.

Remark 7.3. This definition is weaker than claiming a unique canonical theory on $\mathbb{H}_K$, but it is mathematically meaningful. It is also close to how continuum quantum field theories are usually defined: by regularization, counterterms, and a limiting procedure.

8 Potential-theoretic kernels

The finite-skeleton theory is rigorous. Potential theory supplies natural candidates for continuum kernels.

8.1 Berkovich Green functions

Let μ be a probability measure on $\mathbb{P}_K^{1, \text{an}}$ with continuous potential. The normalized Arakelov–Green function $g_\mu(X, Y)$ is characterized, up to the usual normalization, by

$$\Delta_X g_\mu(X, Y) = \delta_Y - \mu, \quad \int g_\mu(X, Y) d\mu(X) = 0. \quad (18)$$

Here Δ is the Berkovich Laplacian in the sense of non-Archimedean potential theory.

A natural generalized free boundary covariance is

$$C_{\Delta,\mu}(x, y) := N_2(\Delta) \exp(2\Delta g_\mu(x, y)), \quad x, y \in \mathbb{P}^1(K), \quad x \neq y, \quad (19)$$

provided the right-hand side is finite off the diagonal. With the common convention $g_\mu(x, y) = -\log[x, y] + O(1)$, this has the expected power singularity. This is a boundary object. It does not automatically come from a canonical local bulk action on $\mathbb{H}_K$, but it may arise as a limit of Dirichlet-to-Neumann forms for suitable exhaustions.

8.2 Hsia-kernel Poisson ansatz

For $X \in \mathbb{H}_K$ and $x \in \mathbb{P}^1(K)$, define the candidate bulk-to-boundary kernel

$$K_{\Delta,\mu}^{\text{cand}}(X; x) := C_\Delta \exp(2\Delta g_\mu(X, x) - \Delta g_\mu(X, X)). \quad (20)$$

In the affine chart with the Gauss-point normalization, this reduces heuristically to

$$K_\Delta^{\text{cand}}(\zeta_{a,r}; x) \sim C_\Delta \frac{r^\Delta}{\max\{r, |a - x|\}^{2\Delta}}.$$

Warning 8.1. Equation (20) should be read as a candidate continuum kernel. To turn it into a theorem, one must specify the massive operator for which it is harmonic in X , the bulk measure, and the boundary asymptotics. On the Bruhat–Tits skeleton these data are known; on the full Berkovich line they are additional choices.

9 Dynamical backgrounds

One advantage of the Berkovich formulation is that it naturally incorporates arithmetic dynamics. Let

$$f : \mathbb{P}_K^1 \longrightarrow \mathbb{P}_K^1$$

be a rational map of degree at least two. Non-Archimedean dynamics associates to f a canonical invariant probability measure μ_f on $\mathbb{P}_K^{1,\text{an}}$, often supported on the Berkovich Julia set.

Given such μ_f , one may define a dynamical-background generalized free covariance by

$$C_{\Delta,f}(x, y) := N_2(\Delta) \exp(2\Delta g_{\mu_f}(x, y)). \quad (21)$$

This is a precise boundary construction. A corresponding bulk dual is obtained only after choosing skeleta adapted to $\text{supp}(\mu_f)$ and proving convergence of the corresponding renormalized Dirichlet-to-Neumann forms.

Proposition 9.1 (Safe dynamical statement). *For every rational map f of degree at least two for which the canonical measure μ_f and Green function g_{μ_f} are defined, formula (21) defines a potential-theoretic candidate for a boundary two-point kernel. It is a Berkovich holographic background precisely when there exists a Berkovich holographic datum $\mathfrak{D}_f$ whose renormalized two-point function equals (21).*

Proof. The first assertion is a definition using the potential-theoretic Green function. The second assertion is the definition of being realized by the finite-skeleton holographic limit (17). $\square$

10 What is proved and what remains conjectural

The construction separates theorem from ansatz.

10.1 Proved finite-skeleton statements

For every finite Berkovich skeleton Γ with boundary B and $m^2 \geq 0$:

- (i) the Dirichlet problem is well-posed;
- (ii) the Poisson operator $H_{\Gamma,m}$ exists and is unique;
- (iii) the finite bulk-to-boundary kernels $K_{\Gamma,m}(X;b)$ are well-defined;
- (iv) the on-shell free action is the Dirichlet-to-Neumann form;
- (v) tree-level contact diagrams are finite integrals over Γ ;
- (vi) the Bruhat–Tits scalar theory is recovered by taking graph skeleta and counting measure.

10.2 Continuum statements requiring extra work

The following claims are not automatic and should be treated as problems or conjectures:

- (i) existence of a canonical bulk measure on all of $\mathbb{H}_K$;
- (ii) existence of a canonical massive Berkovich Laplacian whose Green kernel is $e^{-\Delta\rho(X,Y)}$;
- (iii) a universal Berkovich mass-dimension formula independent of skeleton choices;
- (iv) convergence of renormalized finite-skeleton Dirichlet-to-Neumann forms;
- (v) realization of arbitrary potential-theoretic covariances $e^{2\Delta g_\mu(x,y)}$ by local bulk actions.

10.3 Summary of the corrected theory

The correct Berkovich enlargement of p -adic AdS/CFT is the following.

A Berkovich holographic scalar theory is not defined merely by declaring the bulk to be $\mathbb{H}_K$. It is defined by a system of finite metric subtrees of $\mathbb{P}_K^{1,\text{an}}$, each carrying a well-posed scalar Dirichlet problem, together with a renormalized exhaustion limit toward the type-I boundary. The Bruhat–Tits theory is the homogeneous regular-tree case. The Hsia kernel and Berkovich Green functions provide natural continuum candidates for propagators and boundary correlators, but their bulk interpretation requires the finite-skeleton limiting construction.

Symbolically, the correct replacement is

$$\text{Bruhat–Tits tree model} \rightsquigarrow \left\{ \begin{array}{l} \text{finite Berkovich skeleta} \\ \text{Dirichlet-to-Neumann maps} \\ \text{renormalized exhaustion limit} \end{array} \right\}.$$

This formulation is rigorous at finite cutoff and flexible enough to include both the standard homogeneous p -adic model and nonhomogeneous potential-theoretic backgrounds from Berkovich dynamics.

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